From Queue to Capital

AI-driven portfolio management for an integrated U.S. energy book: 20 offshore Gulf lease redevelopments and 20 ERCOT queue requests under one budget and 2,000 joint scenarios

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github.com/alanmossinger/Capstone

The decision

An energy company holds a budget of $6.15B (40% of forty candidates' capital), no technology above 40% of it, over 2025–2044. The candidates: the 20 largest producing federal offshore Gulf of Mexico leases (BSEE, 2010–2024), to be redeveloped; and 20 ERCOT power requests in the interconnection queue (5 gas combined cycle, 5 solar, 5 wind, 5 storage) that may never reach operation. The queue-survival models of Part I are the timing engine of the power side.

Four questions, criteria registered before scoring. RQ1 can machine learning improve the physical forecasts (C20)? RQ2 how do forecast error, price uncertainty and correlation change risk? RQ3 does a risk-aware book beat NPV ranking and sector-only books out of sample (C21–C24)? RQ4 which constraints change the selection?

What each candidate is worth

Under the confirmed upstream cost grid (ASSUMED, shown before it is used) the twenty leases carry **$14.6B** of incremental NPV at base, every one positive, median break-even **$33 a barrel** against a 2024 real WTI of $77. The twenty requests are worth **–$0.76B merchant** and **–$0.28B contracted** once the queue models' chance of operating within seven years (6%–23%) and the cost to carry are priced in; only wind under a $62 PPA is positive.

The book's $13.8B expected NPV moves most with uplift_share and opex_usd_per_boe; every bar is an assumption, not a measurement

One-at-a-time sensitivity of the default book (20 leases + 20 ERCOT requests) to the upstream cost grid of decision #68 and to the 2024 price basis of the power candidates.

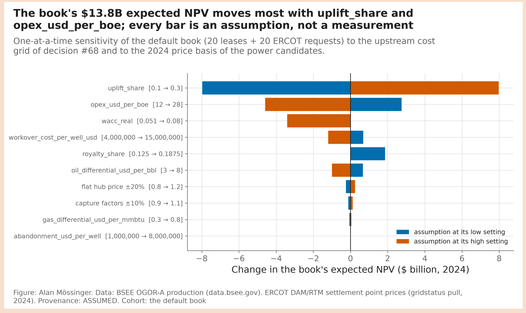


Figure: Alan Mössinger. Data: BSEE OGOR-A production (data.bsee.gov). ERCOT DAM/RTM settlement point prices (gridstatus pull, 2024). Provenance: ASSUMED. Cohort: the default book

The uplift share (10%–30%: $5.9B to $21.8B) and the operating cost ($12–$28 per boe: $16.6B to $9.2B) decide the lease side; the power price basis moves the book by under $0.5B.

The four books under one budget — C21–C24 fail, and why

Out of sample the risk-aware book earns $20.7B with a worst-5% outcome that is still a $0.3B profit, against $21.1B and a worst-5% outcome that is still a $0.3B profit for top-N NPV; C21 FAIL, C22 FAIL, C23 FAIL, C24 FAIL

Left: the four books and the two single-sector books on the held-out copula scenarios, with the MILP frontier over λ. Right: capital by technology in each book; budget $6.15B, no technology above 40%. Contracted revenue case.

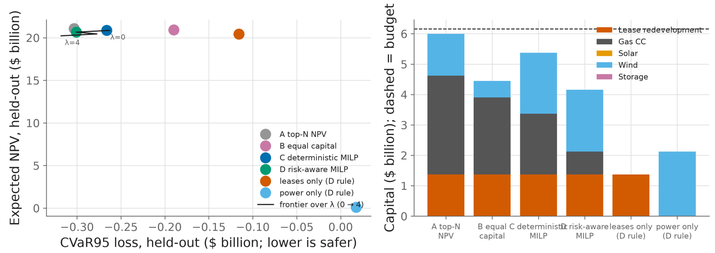


Figure: Alan Mössinger. Data: BSEE OGOR-A production (data.bsee.gov). ERCOT DAM/RTM settlement point prices (gridstatus pull, 2024). Provenance: FORECAST. Cohort: the forty candidates of notebook 07; scenarios of notebook 08

A top-N NPV · B equal capital by sector · C deterministic MILP · D risk-aware MILP (E[NPV] – λ·CVaR95, HiGHS), selected on block-bootstrap scenarios and scored on a held-out copula set. Out of sample D earns **$20.65B** against **$21.07B** for A (C21: −$420M [−$659M, −$210M]); no more than B (C22); its selection moves more than C's under six stresses (Jaccard 0.95 vs 0.99, C23); its tail is not below the single-sector comparator's (C24). **The finding:** no book of these forty loses money in any of 2,000 joint scenarios under the confirmed grid, so risk aversion has nothing to buy — a statement about the grid and the prices, not the method.

The decision gym: AI and rule against instinct, scored on outcomes

Across 16 past situations the registered rule beats instinct by $+100M a situation on average; the advisor's picks are not yet recorded

Top: queue situations as of their date (realized value = modeled value per MW × realized operation − cost to the gate, a DERIVED proxy); in 9 of them none of the fifteen largest requests operated within the window, so the best book is empty. Bottom: the lease regimes, scored on the realized price path. The tick is the best book with hindsight.

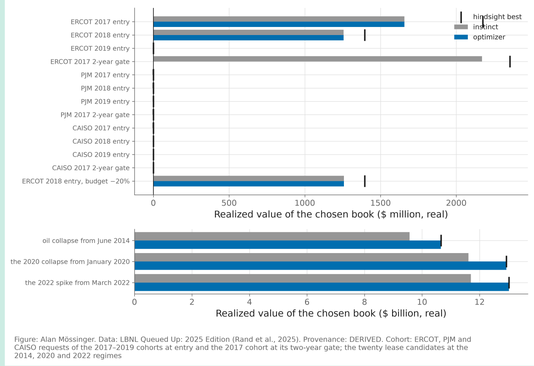


Figure: Alan Mössinger. Data: LBNL Queued Up: 2025 Edition (Rand et al., 2025). Provenance: DERIVED. Cohort: ERCOT, PJM and CAISO requests of the 2017-2019 cohorts at entry and the 2017 cohort at its two-year gate; the twenty lease candidates at the 2014, 2020 and 2022 regimes

Sixteen situations built as of their date with no hindsight: the fifteen largest requests of ERCOT, PJM and CAISO in each 2017–2019 cohort at entry, the 2017 cohort at its two-year gate, a 2018 budget cut, and the lease candidates at the 2014, 2020 and 2022 oil-price regimes. The registered rule beat instinct by **$100M a situation** (regret against hindsight $197M vs $297M); in nine queue rooms none of the largest requests operated, and the rule's edge is the money it did not spend.

The AI inside the decision

The advisor sees only the result tables, every number it writes is checked, every call is recorded and replayed, and the human overrides only through the gate

Grey: data and records. Blue: the prompts and the check. Orange: the model and the gym. Green: the human and the gate. Arrows are the flow of one decision.

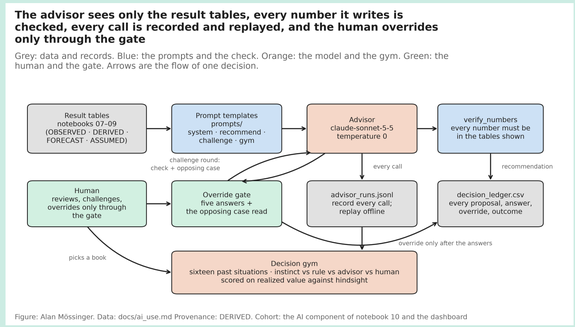


Figure: Alan Mössinger. Data: docs/ai_use.md Provenance: DERIVED. Cohort: the AI component of notebook 10 and the dashboard

An LLM advisor (Anthropic Messages API, temperature 0) sees only the result tables; every number it writes is checked against them; every call is recorded and replayed, so the pipeline runs offline. It recommends per scenario and horizon, runs the motivated-reasoning check and argues the opposing book (ideological Turing test); the human overrides only after answering the five questions. Every decision goes to a ledger; the dashboard (`make app`) carries both modes.

When: the book at one to twenty years

The lease book turns positive within 3 years; the requests are a cost at every horizon, $0.84B of expected loss by year 20

Left: expected NPV of all forty candidates counted within each horizon, by sector (requests weighted by the queue models' probability of operating by each year). Right: the composition of the deterministic book at each horizon under the registered budget and cap.

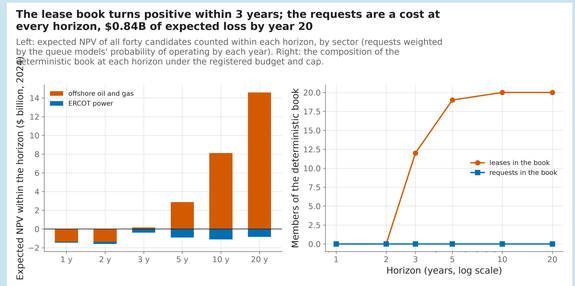


Figure: Alan Mössinger. Data: LBNL Queued Up: 2025 Edition (Rand et al., 2025). Provenance: FORECAST. Cohort: the forty candidates of notebook 07 at the flat 2024 real prices

The lease book is −$1.4B within two years (the workovers), $0.2B at three, $14.6B at twenty; a request is a carry cost until it operates (−$0.84B at twenty). A one-year owner funds nothing; a three-year owner funds twelve leases; the twenty-year owner buys the contracted wind.

Recommendation and record

Fund the twenty leases and the contracted wind; hold gas to its gate; no solar or storage at 2024 prices — book D: 24 candidates, $4.2B, P(NPV < 0) 1.8%, the worst 5% still a profit. What would change it: the uplift share and the operating cost (ASSUMED); the hub price and the PPA on the power side. Registered record: C1–C5 8/12 · C6–C12 3/12 · C13–C14 3/12 · C15–C17 4/10 · C18–C19 fail · C20 3/12 · C21–C24 0/4. Every number traces to `results/`; every figure carries its provenance.

Data: BSEE OGOR-A; EIA-923 and EIA spot prices; ERCOT settlement-point prices; EPA eGRID; NREL ATB; LBNL Queued Up 2025 (Rand et al., 2025). Author's analysis.